



INF212

ALGORITHMS AND PROGRAMMING II

PROJECT 2

Checkers (Dama) Game



Pointer & Array Passing & Memory

Deadline

20.04.2026 – 23:59

Late submissions will **NOT be accepted.**

1. PROJECT DESCRIPTION

In this project, students are required to develop a 2-player Checkers (Draughts) game using the C programming language.

The main objective of this project is to evaluate the following concepts:

- Pointer usage
- Dynamic memory allocation (malloc / free)
- Passing arrays to functions using pointers
- Modular programming with functions

2. STRICT RULES

- Use of "struct" is forbidden.
- Global variables are NOT allowed.
- Copying code from external sources is forbidden.
- The game board must be created using **dynamic memory allocation**.
- All operations must be implemented using **functions**.
- Arrays must be passed to functions using **pointers**.
- Allocated memory must be released using **free()**.
- You are strictly limited to the programming topics taught in class so far. The use of any built-in function is not allowed. You must implement your own code as explained.
- While implementing your code, divide your work into functions and use these functions properly.
- Do not use any parts of code that you cannot explain, justify, or fully understand. Avoid using code generated directly by LLM models (such as ChatGPT, DeepSeek, Llama, Mistral etc.) if you do not know exactly what it does and how it works.

3. GAME RULES

Board : 8x8 grid

Pieces : Player 1 → 1, Player 2 → 2, Empty cell → 0

Initial Setup : First 3 rows → Player 1, Last 3 rows → Player 2

Movement Rules:

- Pieces move diagonally
- Only forward movement is allowed
- Capturing is performed by jumping over an opponent piece
- Backward movement is NOT allowed

Winning Condition:

- A player wins when the opponent has no remaining pieces

4. FINAL NOTES

- The length of the code does not matter, but it must follow the given rules.
- Incorrect algorithms will receive lower scores
- You are allowed to use examples from the internet to help you complete your project. However, plagiarism is strictly forbidden. Any homework that is found to be an exact copy of another student's work, or copy-pasted from an online source without proper understanding and modification, will receive a grade of 0. Necessary disciplinary actions will be taken according to the regulations.
- You must not use any code generated directly by AI Models — such as ChatGPT, DeepSeek, Llama, Mistral etc. — unless you fully understand what it does, how it works, and can explain it in detail. Using any part of the code that you cannot explain or do not understand is strictly prohibited.
- Please include meaningful comment lines in your code to explain what you are doing in each part of your implementation. Clear code documentation is mandatory.
- Any student who fails to explain their own code during the lab session may lose points or receive a grade of 0 even if the code is working correctly.